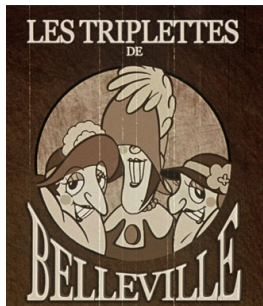


CS319: Scientific Computing**Week 11: Sparse
Matrices and
templates**

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Slides and examples:

<https://www.niallmadden.ie/2425-CS319>

0. Outline



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1. News and Updates

- ▶ Grades for Lab 4, Class Test, and Lab 6 will be posted soon (honest!).
- ▶ Presentations have been scheduled (switch to <https://www.niallmadden.ie/2425-CS319/2425-CS319-Projects.pdf>)

2. Sparse Matrices

Last week we designed a class for representing a matrix. Although we didn't discuss it at the time, the matrices represented are called “**dense**” or “**full**”.

Today we want to see how to store **SPARSE MATRICES**: these are matrices that have so many zeros that it is worth our while exploiting the fact.

2. Sparse Matrices

There are numerous examples of sparse matrices. For example, they occur frequently when solving differential equations numerically.

But perhaps the most obvious example is when we use matrices to represent graphs and **networks**.

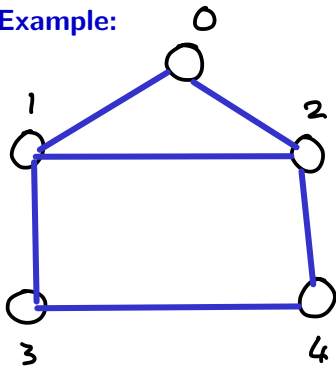
Most real world networks have far more nodes/vertices than they do connections/edges between those nodes.

In a computational setting, most graphs/networks are represented as a matrix, such as the **adjacency matrix**.

- ▶ If the graph has N vertices, the matrix, A , has N rows and columns: each corresponds to vertex.
- ▶ If (i, j) is an edge in the graph, then $a_{ij} = 1$. Otherwise, $a_{ij} = 0$.

2. Sparse Matrices

Example:



	0	1	2	3	4
0	0	1	1	0	0
1	1	0	1	1	0
2	1	1	0	0	1
3	0	1	0	0	1
4	0	0	1	0	0

Note the matrix has 25 entries.
which is quite a lot, given we've
only 6 edges. This has $NNZ: 12$.

2. Sparse Matrices

Compared to the over-all number of entries in the matrix, the **number of non-zeros (NNZs)** is relatively small. So it does not make sense to store them all. Instead, one uses one of the following formats:

- ▶ **Triplet** (which we'll look at presently),
- ▶ **Compressed Sparse Row** storage (CSR) (after triplet)
- ▶ **Compressed Sparse Column** storage (CSC)

And the following formats for very specialised matrices, which we won't study in CS319:

- ▶ **Block Compressed Row/Column Storage**
- ▶ **Compressed Diagonal Storage**
- ▶ **Skyline**

Although the representation and manipulation of sparse matrices is an major topic in Scientific Computing, there isn't a universally agreed definition of an (abstract) *sparse matrix*.

This is because, when coding, we should ask the question: **“When is it worth the effort to store a matrix in a sparse format, rather than in standard (dense) format?”**

The answer is often context-dependent. But roughly, use a sparse format when

- ▶ The memory required by the sparse format is less then the “dense” (or “full”) one;
- ▶ The expense of updating the sparse format is not excessive;
- ▶ Computing a **MatVec** is faster for a sparse matrix.

For **triplet** form we store a **sparse** matrix with **NNZ** non-zeros in three arrays **I** :

- ▶ an **unsigned integer** arrays **I[NNZ]**, with row indices
- ▶ an **unsigned integer** arrays **J[NNZ]**, with column indices.
- ▶ a **double** array **X[NNZ]**.
- ▶ Then entry a_{ij} is stored as $I[k]=i$, $J[k]=j$, $X[k]=a_{ij}$, for some k .

This is also known as **Coordinate list (COO)**

NNZ = "number of non-zeros".

Example: write down the triplet form of the following matrix:

$$\begin{array}{cccc}
 0 & 1 & 0 & 11 & 0 \\
 1 & 1 & 0 & 2 & 0 \\
 2 & 9 & 19 & 0 & 29 \\
 3 & 0 & 0 & 0 & 5 \\
 0 & 1 & 2 & 3 &
 \end{array}$$

$$I = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \\ 2 \\ 2 \\ 3 \\ 3 \end{bmatrix}$$

$$J = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \\ 0 \\ 1 \\ 3 \\ 3 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 \\ 11 \\ 1 \\ 2 \\ 9 \\ 19 \\ 29 \\ 5 \end{bmatrix}$$

$$NNZ = 8$$

In Compressed Row (see later) replace
 I with $R = [0, 2, 4, 7, 8]$
 : $\uparrow_{NNZ}$

Example: Suppose A is a square $N \times N$ matrix, with $N = 1000$, three non-zero entries per row, and all data stored as `doubles`.

- (a) How many values need to be stored if the A is stored in the usual “dense” format? How much memory is required?
- (b) How many values need to be stored if the A is stored in **triplet** format? How much memory is required?

We have $N^2 = 1,000,000$ values to store. Each requires 8 bytes. Total is 8,000,000 bytes, which is roughly 8 Mb.

NNZ is 3,000. So we store 9,000 values (3,000 for each of I, J, X).
 I, J store ints, which need 4 bytes each. They need 12,000 bytes. X needs 24,000. Total 36,000 ≈ 36 Kb.

Our next goal is implement a triplet matrix as a `class`. The main tasks are:

- ▶ Decide what private data elements are needed.
- ▶ Decide what public methods are needed.
- ▶ Implement a matrix-vector multiplication algorithm.

Discussion...

DATA

Arrays for I, J, X .
 N (size of matrix).
 NNZ .

METHODS

- Same as `matrix`.
- Also:
"where" - gives
 k value for i, j .
Also get i, j, x for
given k .

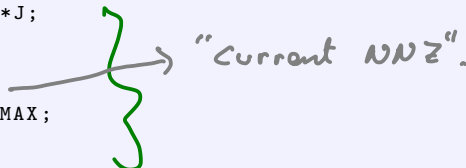
3. Coding triplet

Triplet.h

```
1 // Triplet.h: For 2425-CS319 Week 11
  // Author: Niall Madden
3
4 #ifndef _TRIPLET_H_INCLUDED
5 #define _TRIPLET_H_INCLUDED
6
7 #include "Vector10.h"
  #include "Matrix11.h"
```

3. Coding triplet

Triplet.h

```
10 class Triplet {  
    friend Triplet full2Triplet(Matrix &F, unsigned NNZ_MAX);  
12 private: int full2Triplet  
    unsigned *I, *J;  
14 double *X;  
    unsigned N;  
16 unsigned NNZ;  "Current NNZ".  
    unsigned NNZ_MAX;  
  
public:  
20 Triplet (unsigned N, unsigned nnz_max); // Constructor  
    Triplet (const Triplet &t); // Copy constructor  
22 ~Triplet(void);  
  
24 Triplet &operator=(const Triplet &B); // overload assignment
```

3. Coding triplet

Triplet.h

```
26 unsigned size(void) {return (N);};  
   int where(unsigned i, unsigned j); // negative return on error  
28 unsigned nnz(void) {return (NNZ);};  
   unsigned nnz_max(void) {return (NNZ_MAX);};  
  
   double getij (unsigned i, unsigned j);  
32 void setij (unsigned i, unsigned j, double x);  
  
34 unsigned getI (unsigned k) { return I[k];};  
   unsigned getJ (unsigned k) { return J[k];};  
36 double getX (unsigned k) { return X[k];};  
  
38 Vector operator*(Vector u);  
   void print(void);  
40 };  
#endif
```

→ compute a Triplet - vector product.

3. Coding triplet

Triplet.cpp

```
1 // Triplet.cpp for 2425-CS319 Week 11
  // What: Methods for the Triplet class
3 // Author: Niall Madden
  #include <iostream>
5 #include <iomanip>
  #include "Vector10.h"
7 #include "Matrix11.h"
  #include "Triplet.h"
```


3. Coding triplet

Triplet.cpp (Constructor)

```
10 // Standard constructor.
11 Triplet::Triplet (unsigned int N, unsigned nnz_max) {
12     this->N = N;
13     this->NNZ_MAX = nnz_max;
14     this->NNZ = 0;
15
16     X = new double [nnz_max];
17     I = new unsigned [nnz_max];
18     J = new unsigned [nnz_max];
19     for (unsigned k=0; k<nnz_max; k++) {
20         I[k]=-1;
21         J[k]=-1;
22         X[k]=(double) NULL;
23     }
24 }
```

When the
Matrix
is
created
it is
empty.

Represent that by setting $I[k] = J[k] = -1$.

3. Coding `Triplet`

When using a `Triplet` object to represent a matrix, T , we often need to find where in the array X , the value of t_{ij} is stored. That is done by the following function.

`Triplet.cpp` (where)

```
56 int Triplet::where(unsigned i, unsigned j)
   {
       unsigned int k=0;
58     do {
           if ( (I[k]==i) && (J[k]==j) )
60         return(k);
           k++;
62     } while (k<NNZ);
       return(-1);
64 }
```

[Finished here Friday]

3. Coding triplet

Triplet.cpp (setij)

```
68 void Triplet::setij (unsigned i, unsigned j, double x)
69 {
70     if (i>N-1)
71         std::cerr << "Triplet::setij(): i Index out of bounds." << std::endl;
72     else if (j>N-1)
73         std::cerr << "Triplet::setij(): j Index out of bounds." << std::endl;
74     else if (NNZ > NNZ_MAX-1)
75         std::cerr << "Triplet::setij(): Matrix full." << std::endl;
76     else
77     {
78         int k=where(i,j);
79         if (k == -1) // nothing stored in [i,j].
80             {
81                 I[NNZ]=i;
82                 J[NNZ]=j;
83                 X[NNZ]=x;
84                 NNZ++;
85             }
86         else
87             X[k]=x;
88     }
```

3. Coding triplet

Triplet.cpp (operator *)

```
180 Vector Triplet::operator*(Vector u)
181 {
182     Vector v(N); // v = A*u, where A is the implicitly passed Triplet
183     v.zero();
184     if (N != u.size())
185         std::cerr << "Error: Triplet::operator* - dimension mismatch"
186                     << std::endl;
187     else
188         for (unsigned k=0; k<NNZ; k++)
189             v.seti(I[k], v.geti(I[k]) + X[k]*u.geti(J[k]));
190     return(v);
191 }
```

3. Coding triplet

To demonstrate the use of the `Triplet` class, I've included a program called `00triplet_example.cpp` which shows how to use the Jacobi method to solve a linear system where the matrix is stored in triplet format.

It also provides a (better) way of timing code, and gives the speed-up achieved for given parameters.

You can also run that programme directly on [`online-cpp.com`](https://www.online-cpp.com/):
[`https://www.online-cpp.com/HzT3v4N7t`](https://www.online-cpp.com/HzT3v4N7t)



See link on
website.

4. Compress Sparse Row

If we know that the entries in our matrix are stored in order, then it is possible to store the matrix more efficiently than in Triplet format. One way of doing this is to use **CSR: Compressed Sparse Row**, also known as *Yale Format*.

The matrix is stored in 3 vectors:

- ▶ a `double` array, `x` of length `nnz` (“number of nonzero entries”) storing the non-zero entries matrix, in column-wise order.
- ▶ an `int` array, `c` of length `nnz` storing `column` index of the entries. That is, `x[k]` would be found in column `c[k]` of the full matrix.
- ▶ an `int` array, `r` of length `N + 1`, where `r[i]` stores that **starting point** of row `i` as it appears in the arrays `x` and `c`, and `r[N] = nnz`.

Note: this format is used by default in Python by the `scipy.sparse` module (also `networkx`, `scikit-learn`, ...).

4. Compress Sparse Row

Example

Show how the matrix below would be stored in CSR

$$\begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 2 & -1 & 0 & 0 \\ 0 & 5 & -1 & 0 \\ 0 & -2 & 4 & 0 \\ -3 & 0 & 0 & 4 \end{pmatrix} \end{matrix}$$

$$x = [2, -1, 5, -1, -2, 4, -3, 4]$$

$$c = [0, 1, 1, 2, 1, 2, 0, 3]$$

$$r = [0, 2, 4, 6, 8]$$

The process of multiplying a matrix (in CSR) by a vector is rather simple:

```
int index=0;
2 for (int row=0; row<N; row++)
    for (i=r[row]; i<r[row+1]; i++)
4     {
        j=c[index];
6         v[j] += x[index]*b[i];
        index++;
8     }
```

.....

I don't provide code for implementing a CSR class here: that is an exercise.

5. Templates

We've written several of our own functions and classes. For most of these, they depend on data of a certain **type**. For example, in Week 5, we wrote some functions called `Sort()` for sorting `ints`.

Suppose we wanted to sort arrays of a different type, e.g.,

`strings`. We could take our old

`Sort(int *list, int length)` function from

`Week05/04Sort.cpp` and rewrite it as

`strings: Sort(string *list, int length)`

The code the “new” function would be almost identical: we'd just replace several instances of the datatype `int` with `string`.

To avoid this repetition, and to allow us to write functions or class **generic** datatypes, C++ provides **templates**.

Today we first consider **function templates**. We'll return to the related idea of **class templates** later.

To perform essentially identical operations for different types of data compactly, use function templates.

- ▶ Syntax: `template <typename T>` immediately precedes the function definition. It means that we'll be referring to the generic datatype as `T` in the function definition.
- ▶ Write a single function template definition. In it, the generic datatype is named `T`.
- ▶ Based on the argument types provided in calls to the function, the compiler automatically creates functions to handle each type of call appropriately.

In the example below, which you can find in detail in `01FunctionTemplate.cpp`, we'll write three functions:

- a `PrintList(MyType *x, int n)`
- b `void Sort(MyType &a, MyType &b)`
- c `void Sort(MyType *x, int n)`

The function prototypes:

01FunctionTemplate.cpp

```
14 template <typename MyType>  
void PrintList(MyType *x, unsigned int n);  
  
16 template <typename MyType>  
void Sort(MyType &a, MyType &b);  
  
20 template <typename MyType>  
void Sort(MyType *list, unsigned int length);
```

The (bubble) `Sort` functions:

`01FunctionTemplate.cpp`

```
52 template <typename MyType>
void Sort(MyType &a, MyType &b) {
54     if (a>b)
    {
        MyType tmp=a;
56     a=b;
        b=tmp;
58     }
}
```

Important: this assumes
that the operator `>` is
overloaded for this type.

```
68 template <typename MyType>
void Sort(MyType *x, unsigned int n) {
    for (int i=n-1; i>1; i--)
70     for (int k=0; k<i; k++)
        Sort(x[k], x[k+1]);
72 }
```

01FunctionTemplate.cpp

```
22 int main(void )  
   {  
24     int Numbers[8];  
     char Letters[8];  
  
     for (int i=0; i<8; i++)  
28       Numbers[i]=rand()%40;  
  
30     for (int i=0; i<8; i++)  
       Letters[i]='A'+rand()%26;
```

01FunctionTemplate.cpp

```
34  std::cout << "Before sorting:" << std::endl;
    std::cout << "Numbers: ";  PrintList(Numbers, 8);
    std::cout << "Letters: ";  PrintList(Letters, 8);

    Sort(Numbers, 8);
38  Sort(Letters, 8);

40  std::cout << "After sorting: " << std::endl;
    std::cout << "Numbers: ";  PrintList(Numbers, 8);
42  std::cout << "Letters: ";  PrintList(Letters, 8);
```

Typical output

Before sorting:

Numbers: 23 6 17 35 33 15 26 12

Letters: B H C D A R Z 0

After sorting:

Numbers: 6 12 15 17 23 26 33 35

Letters: A B C D H 0 R Z

6. The Standard Template Library

During the semester, we've focused on designing classes that can be used to solve problems. These included classes: `Stack`, `Vector` and `Matrix`.

However, most of you worked out that, to some extent, these are already supported in C++. The motivations for reinventing them included

- ▶ our implementation is simple to use;
- ▶ we learned important aspects of C++/OOP;
- ▶ we needed to achieve specific tasks efficiently: this is particularly true of our design of sparse matrix classes.

Now we'll look at how to use the built-in implementation that comes with the C++ **Standard Template Library (STL)**.

The **STL** provides

- (1) **Containers:** ways of collecting/storing items of some type (template....)
- (2) **Iterators:** for accessing items in the containers
- (3) **Algorithms:** for operating on the contents of containers, such as finding a particular item, or sorting (a subset) of them.
- (4) **functors:** essentially, a class which defines the operator(). We won't say more than this right now.

It has to be noted, though: the STL is not that easy to use. In particular the error messages generated are rather verbose and unhelpful.

A **container** stores objects/elements. These elements can have basic data-type (e.g., `char`, `int`, `double`, ...) or can be objects (e.g., `string`, or user-defined objects).

The most important types of containers are:

vector: an indexed sequence (often called “*random access*”, though this would be better called “*arbitrary access*”). All the items are of the same type. It can be resized, and have new items added to the end. One can also add items to positions not the end, but this is slow.

`set`: a collection of unique items (of the same type), stored in order. When defined relative to a user-defined class, an overloaded `operator<` (less than) must be provided for correct operation.

`mset`: an ordered collection, like a set, but can have repeated values.

`list`: a doubly linked list.

`stack`: a stack.

... etc...

We'll focus on `sets`, `msets` and `vectors`.

An **iterator** is an object used to select (or move between) elements in a container.

We can think of them as pointers, that allow us to reference particular elements.

They come in particular flavours:

- ▶ forward, reverse, and bidirectional iterators;
- ▶ random-access/indexed-access iterators;
- ▶ input and output iterators;

7. sets and multisets

To use a `set` or `multiset`, we must

```
#include <set>
```

Suppose we want to create a `multiset` to store `strings` (which just happen to be passwords...), and an iterator for it, we could define

```
1 std::multiset<std::string> multi_pwd;  
   std::multiset<std::string>::iterator multi_pwd_i;
```

note: we are really using templates.

To add an item to the (multi)set, we could use

```
multi_pwd.insert(MyString);
```

This will add the new string to the `multiset`, automatically choosing its position so that it remains ordered. (If we use a `set`, it gets inserted into the correct position, providing this does not result in duplication).

Finished here Friday @ 12.

7. sets and multisets

Other important methods include

- ▶ `begin()` (returns an iterator that points to the first element)
- ▶ `end()` (returns an iterator that points to *one past* the end of the set).
- ▶ `clear()` (remove contents)
- ▶ `count()` (count number of occurrences)
- ▶ `empty()` (is the set empty?)
- ▶ `erase()` (remove an element, or range of elements)
- ▶ `find()` (locate an element; return an iterator)
- ▶ `size()` (number of elements)
- ▶ `swap()` (swap contents of two sets of same type)
- ▶ `for_each()` (apply a particular function to each item in a container)

7. sets and multisets

An example of using `begin` and `end` with a set and `mset`:

02set_and_multiset.cpp

```
10 int main(void )
11 {
12     std::set <int> set_int;
13     std::set <int>::iterator set_int_i;
14     std::multiset <int> multi_int;
15     std::multiset <int>::iterator multi_int_i;
16
17     for (int i=0; i<=20; i+=3) // (0,3,6,9,12,15,18)
18     {
19         set_int.insert(i);
20         multi_int.insert(i);
21     }
22     for (int i=20; i>0; i-=2) // (20,18,16,...,4,2)
23     {
24         set_int.insert(i);
25         multi_int.insert(i);
26     }
```

7. sets and multisets

First, we will see how to iterate over the `multiset`:

02set_and_multiset.cpp

```
28  std::cout << "The multiset has " << multi_int.size() <<
    " items." << std::endl;
30  std::cout << "\t They are: ";
    for (multi_int_i = multi_int.begin();
32      multi_int_i != multi_int.end();
        multi_int_i++)
34      std::cout << std::setw(3) << *multi_int_i;
    std::cout << std::endl;
36  std::cout << "\t 6 occurs " << multi_int.count(6) <<
    " time(s)." << std::endl;
```

The output is

```
1 The multiset has 17 items.
   They are:  0  2  3  4  6  6  8  9 10 12 12 14 15 16 18 18 20
3   6 occurs 2 times.
```

7. sets and multisets

Next we will iterate over the `set`:

02set_and_multiset.cpp

```
40  std::cout << "The set has " << set_int.size() <<
    " items." << std::endl;
    std::cout << "\t They are: ";
42  for (set_int_i = set_int.begin();
        set_int_i != set_int.end();
44      set_int_i++)
        std::cout << std::setw(3) << *set_int_i;
46  std::cout << std::endl << "\t 6 occurs " << set_int.count(6)
        << " time(s)." << std::endl;
```

The output is

```
1  The set has 14 items.
   They are:   0  2  3  4  6  8  9 10 12 14 15 16 18 20
3  6 occurs 1 time.
```


8. vector

To use `vector`, we must

```
1 #include <vector>
```

Unlike a set, we can access a vector by index. Moreover, by default it is not sorted, though there are algorithms to sort its contents.

Since it is unordered, a new item usually gets added to the end, using `push_back`

This can be removed, using `pop_back`

Other important methods include

- ▶ `at`
- ▶ `operator[]`
- ▶ `back` (not the same as `end`)

8. vector

03STL_vector.cpp

```
10 #include <vector>           // vector
11 #include <algorithm>        // sort
12 void print_int (int i) { std::cout << std::setw(3) << i; }
13 int main(void )
14 {
15     std::vector <int> vec_int;
16     std::vector <int>::iterator vec_int_i;
17     std::cout << "Vector has " << vec_int.size() <<
18         " elements." << std::endl ;
19
20     for (int i=3; i>=0; i--)
21         vec_int.push_back(i*3); // (9,6,3,0)
22
23     std::cout << "Vector has " << vec_int.size() << " elements: ";
24     for (unsigned int i=0; i<vec_int.size(); i++)
25         std::cout << std::setw(3) << vec_int[i];
```

Output (so far):

```
1 Vector has 0 elements.
  Vector has 4 elements:   9   6   3   0
```

8. vector

This snippet demonstrates the use of

- ▶ the `find` and `insert` methods;
- ▶ the `for_each` iterate through an entire container.

03STL_vector.cpp

```
28  vec_int_i = find (vec_int.begin(),vec_int.end(),3);  
    vec_int.insert(vec_int_i,10);  
  
30  std::cout << std::endl;  
    std::cout << "Vector has " << vec_int.size() << " elements: ";  
32  for_each (vec_int.begin(), vec_int.end(), print_int);
```

Output (continued):

```
1  Vector has 0 elements.  
   Vector has 4 elements:    9    6    3    0  
3  Vector has 5 elements:    9    6  10    3    0
```

8. vector

Finally, we show how to `sort` the items in the list:

03STL_vector.h

```
36  std::cout << "Sorting the vector..." << std::endl;  
    sort(vec_int.begin(), vec_int.end());  
    std::cout << "Now vector is: ";  
38  for_each (vec_int.begin(), vec_int.end(), print_int);
```

Output (all):

```
1  Vector has 0 elements.  
   Vector has 4 elements:    9    6    3    0  
3  Vector has 5 elements:    9    6  10    3    0  
   Sorting the vector...  
5  Now vector is:    0    3    6    9  10
```

Other important methods include

- ▶ `begin()` (returns an iterator that points to the first element)
- ▶ `end()` (returns an iterator that points to *one past* the end of the set).
- ▶ `clear()` (remove contents)
- ▶ `count()` (count number of occurrences)
- ▶ `empty()` (is the set empty?)
- ▶ `erase()` (remove an element, or range of elements)
- ▶ `find()` (locate an element; return an iterator)
- ▶ `size()` (number of elements)
- ▶ `swap()` (swap contents of two sets of same type)
- ▶ `for_each()` (apply a particular function to each item in a container)

The **ranged-based** *for* loop is a recent addition to C++, so it might not work with old compilers. With g++, you may need to enable the `c++11` option.

In the code above, the line

```
1 for_each (vec_int.begin(), vec_int.end(), print_int);
```

could be replaced with

```
1 for (int i : vec_int)
    print_int(i);
```

9. Algorithm

To use `algorithm`, we must

```
#include <algorithm>
```

Useful functions that this provides include

- ▶ `for_each`
- ▶ `sort` and `partial_sort`
- ▶ `search`
- ▶ `copy` and `fill`
- ▶ `merge`
- ▶ `set_union`, `set_difference`
- ▶ etc.